



OPEN ACCESS

EDITED BY
Gabriele Giorgi,
European University of Rome, Italy

REVIEWED BY
Tahrima Ferdous,
Queensland University of
Technology, Australia
Tengku Rika Valentina,
Andalas University, Indonesia

*CORRESPONDENCE
Upma Gautam
✉ upma@ipu.ac.in

RECEIVED 24 March 2026
REVISED 01 July 2026
ACCEPTED 07 July 2026
PUBLISHED 27 July 2026

CITATION
Bajpai Tewari D, Gautam U, Mishra A,
Verma K, Jaiswal V and Das P (2026)
Strategizing safety for women in the gig
economy: a scoping review of policy
research.
Front. Organ. Psychol. 4:1838133.
doi: 10.3389/forgp.2026.1838133

COPYRIGHT
© 2026 Bajpai Tewari, Gautam, Mishra,
Verma, Jaiswal and Das. This is an
open-access article distributed under the
terms of the [Creative Commons
Attribution License \(CC BY\)](https://creativecommons.org/licenses/by/4.0/). The use,
distribution or reproduction in other
forums is permitted, provided the
original author(s) and the copyright
owner(s) are credited and that the
original publication in this journal is
cited, in accordance with accepted
academic practice. No use, distribution
or reproduction is permitted which does
not comply with these terms.

Strategizing safety for women in the gig economy: a scoping review of policy research

Deeksha Bajpai Tewari¹, Upma Gautam^{2*}, Ankita Mishra²,
Kanishka Verma³, Vaishnavi Jaiswal² and Priya Das⁴

¹Department of Geography, Dyal Singh College, University of Delhi, New Delhi, India, ²University School of law and Legal Studies, Guru Gobind Singh Indraprastha University, New Delhi, India, ³Department of Geography, Delhi School of Economics, University of Delhi, New Delhi, India, ⁴Campus Law Centre, Faculty of Law, University of Delhi, New Delhi, India

Introduction: The gig economy has expanded significantly in urban public spaces; however, the perception of unsafe urban spaces by women and their families restricts their participation in the gig economy. This study aims to identify the safety concerns of female gig workers and the strategies adopted by various stakeholders to address them.

Methods: An advanced search was performed in online databases, including Web of Science, Scopus, and APA PsycInfo, to identify articles pertaining to the safety of female gig workers for this scoping review. PRISMA approach for scoping review (PRISMA-ScR) was employed to assess the scope of relevant literature and synthesize the prevalent safety concerns and strategies adopted by various stakeholder to mitigate that concern. A narrative and tabular synthesis approach was used to describe the collected data.

Results: A total of 482 studies were identified and retrieved from the databases. Sixteen studies were eligible for inclusion in this review. Abusive behavior in the workplace, fear of nighttime operations, working on digital platforms, customer anonymity, and lack of client verification were identified as prominent safety concerns. Apart from scant informal strategies, there are no formal structures that address harassment against female gig workers.

Conclusion: Several barriers pose safety concerns for female gig workers, thereby hampering their participation in the workforce. An altered approach to the existing legal and policy framework will ensure the safety of female gig workers from sexual harassment at the workplace.

KEYWORDS

gig economy, PRISMA-ScR, safety, strategies, women

Introduction

Digital platforms have become the mainstay of service delivery owing to technological advancements. This phenomenon, known as the platformization of services, has significantly transformed urban dynamics by altering traditional employment roles and creating new work avenues. This evolving task-oriented economic landscape has an interconnected triadic structure comprising gig workers as service providers, consumers, and digital platforms that act as intermediaries. These platforms allocate tasks to workers using algorithms (Banik and Padalkar, 2021; Cyrille et al., 2019). The ecosystem has witnessed prominent expansion due to rising service demands and the proliferation of digital platforms, increasing from 142 in 2010 to 777 in 2021.

Most of these platforms are concentrated in the USA (29%), India (8%), and the UK (5%; ILO, 2021). This increase has significantly penetrated sectors such as taxi or ride-hailing, food delivery, and web-based work (ILO, 2021; Ruyter et al., 2019). Currently, 12% of the global labor force is engaged as gig workers (World Bank, 2023). It has revolutionized the future of work and the reach of the service sector among consumers globally and is projected to be worth \$2,1450 billion by 2033. Undoubtedly, the flexibility afforded by these platforms and lower entry requirements can be contended to be prime factors attracting workers not only in great numbers but also from diverse backgrounds.

The contemporary gig economy can be understood as an extension of traditional temporary or freelance work that has long been present in the unofficial job markets. It has evolved and expanded from its earlier forms in the informal sector and is now more prevalent and visible in the digital age (Ayentimi et al., 2023; Huws et al., 2018; Stanford, 2017). The classification of the gig economy into online, web-based, or location-based categories reflects the diverse nature of modern work arrangements. With the rise of digital platforms and globalization, there has been an increase in online gig work, which includes services like freelance writing, graphic design, and virtual assistance, where workers complete tasks remotely from clients regardless of time and distance. Conversely, there has been an enormous growth of location-based gig work in urban areas, including ride-sharing, food delivery, and on-demand home services, with the advent of smartphone apps and GPS technology enabling real-time matching of service providers to customers in their location. As cities continue to become interconnected and consumers are increasingly wanting convenience, location-based gig work will start to compete on a similar economic scale and workforce participation level as online gig work.

Gender disparity is a persistent feature of traditional workforce systems, and it carries over into the online gig economy. Women make up only 42% of the estimated 163 million online gig workers worldwide (World Bank, 2023). The USA and Ukraine have 41 and 39% of female online gig workers, respectively, whereas India has only 21% (ILO, 2021). Labor market stratification represents Labor market stratification shows up here too, much as it does in traditional workplaces. Women's low participation in the online platform economy mirrors the gender discrimination long observed in brick-and-mortar employment, carried over into the platform setting (Anwar, 2022; Cook et al., 2021; Galperin, 2021; Vyas, 2021). The continued growth of location-based services in urban areas matters for female gig workers because these public spaces are, in effect, their workplace. Since digital platforms mediate work on a task-by-task basis rather than through fixed employment, safe streets and work environments are central to whether women choose to join these platforms at all. The low entry barrier, flexibility, autonomy, and work-life balance these platforms offer to women who are burdened with care and household responsibilities; afforded by these platforms; attract women to join them (Ajaiyeoba, 2024; Churchill and Craig, 2019; Manyika et al., 2016; Tandon and Sekharan, 2022). In addition to catering to the employment avenues of local women, these platforms also attract migrant women who face language barriers, financial distress, and whose paperwork is insufficient to join

traditional employment (Ajaiyeoba, 2024; Hau and Borello, 2024; Newlands, 2024; van Doorn et al., 2023). The safety risks women face in the gig economy are best read through several overlapping theoretical lenses, since no single framework captures the whole picture. When Walby (1990) theory of patriarchy explains gender-based oppression in both public and private life, including women's exclusion from certain occupations and their exposure to harassment at work (Walby, 1990); applied here, it suggests that platform-mediated work tends to repeat patriarchal patterns rather than break from them therefore, women cluster in "feminine" service niches, face customer violence, and lack the institutional protections that formal employees have. According to Social role theory socially constructed ideas about femininity and masculinity shape how vulnerable women are in male-dominated sectors such as ride-hailing and food delivery, and explain why some women adopt "masculine" behaviors as a coping strategy (Eagly et al., 2000). Similarly, Intersectionality theory points out that these risks are not evenly distributed and they intensify where gender intersects with class, race, migration status, and geography, which is why migrant women in particular face compounded vulnerability from language barriers, precarious legal status, and limited access to redress (Collins, 2019; Crenshaw, 1989). The theory of algorithmic management posits that the architecture of platforms creates new types of power through rating systems, non-transparent task allocation and automated penalties, resulting in a situation where workers, primarily women, have to absorb the costs associated with governance decisions made without any input from them (Duggan et al., 2020; Muldoon and Raekstad, 2023). Therefore, the algorithm itself serves as a site for the creation of harm such as penalties for refusal, rewards for silence, or preferential treatment of customer experience over worker experience, whenever it produces an outcome that is harmful to women gig workers' safety. The structure of platform capitalism treats gig platforms as infrastructure designed to externalize the risks associated with business operations including safety and classify gig workers as independent contractors, thus insulating the platform from claims related to the effects of harassment and unsafe working conditions (Graham and Shaw, 2017; Srnicek, 2017). The classification of gig workers as independent contractors is not random; this is a deliberate structural choice intended to insulate the platform from liability for harassment or unsafe working conditions, thereby creating a regulatory vacuum for female gig workers. These theoretical frameworks establish the reasons why women gig workers are at increased risk for unsafe working conditions and the adequacy of responses addressing the risks of women gig workers at the individual, collective, platform, and state level. Extensive global research indicates that gender discrimination and stereotypes, prevalent in traditional employment sectors, have also manifested within the platform economy. This is demonstrated by the concentration of women in gig sectors considered "appropriate" for women, such as beauty services, and their underrepresentation in "masculine" occupations, including automotive repair, construction, and electrical work (Barzilay and Ben-David, 2017; Sarker et al., 2024; Vyas, 2021). Furthermore, emerging challenges have intensified pre-existing issues of gender inequality, particularly through the algorithmic management of digitally mediated tasks and a complete lack of

transparency in task allocation and performance evaluation. These factors exacerbate gender biases against women and introduce new workplace challenges for female gig workers (Duggan et al., 2020; Hill, 2021; Muldoon and Raekstad, 2023; Sun and Li, 2024). These inherent gender biases manifest at every stage of employment in the gig economy, from hiring to work allocation and wage gaps. Gendered inequalities within the gig economy are not confined to specific regions but demonstrate similar patterns across Western and Asian contexts, shaped by comparable structural dynamics. For instance, in the United States, a total earnings gap of 7% has been identified among male and female Uber drivers in 2015–2017, which occurs due to differences in the number of working hours, driving speed, and location choices. Women tend to work more on familiar routes with lower crime rates and non-price surging hours to balance household responsibilities (Cook et al., 2021). Likewise, in remote tasks, women limit their operational hours to evening hours due to caregiving responsibilities, leading to a reinforcement of income disparities (Dong et al., 2024). Similar patterns have also been observed in China, where women earn 85% of what men earn in a month while working in platform-based delivery and ride-hailing services because platform algorithms and social biases operate as forms of implicit discrimination (Dokuka et al., 2022). These recurring patterns reflect that disparities associated with wages and working hours are driven by gender stereotyping of occupations and preconceived notions of male and female characteristics with respect to job suitability, which further perpetuates gender imbalances and occupational segregation in the gig economy (Ajaiyeoba, 2024; Fairwork, 2023; Galperin, 2021; Nazareno, 2025; Shockley et al., 2017). Furthermore, the untraceability of human resources and virtual management of task allocation, performance assessment, wages, or per-task payment in the gig economy have been reported to depict gender inequality and asymmetric power relations between workers, customers, and platforms, with workers being at the receiving end, resulting in their ID blocking or termination (Cook et al., 2021; Foong et al., 2018; Lain, 2024; Levine, 2025; Lin et al., 2023; Barzilay and Ben-David, 2017; Rosenblat and Stark, 2016; Vyas, 2021; Wood, 2019). This indicates that within this new labor market, old challenges and politics of work have not disappeared; they have reappeared in new forms and shapes, which in turn maintain economic inequalities and social hierarchies (Mark and Shaw, 2017). Furthermore, the inadequacy of virtual management in promptly and adequately responding to grievances and distressing situations encountered by female gig workers aggravates safety concerns, which is another significant challenge faced by female gig workers in the workplace (Galán, 2024; Ma et al., 2022). Given the inapplicability or insufficiency of existing legal and policy frameworks to deal with the safety of female gig workers in the dynamic “workplaces” of the gig economy, this study helps identify factors aggravating female gig workers’ perception of workplace safety. This study paves the way for the formulation of an effective legislative and policy framework to address the safety concerns of female gig workers and ensure their higher work participation in the gig economy.

This scoping review addresses a critical research question concerning the existence and utilization of safety and security

policies and strategies at the individual, aggregator, and state levels to ensure the safety of female gig workers. The central research question guiding this review is: What safety and security policies and strategies exist at the individual, aggregator (platform), and state levels to ensure the safety of female gig workers from sexual harassment and assault in platform-mediated work? Existing reviews of gender in the gig economy have mostly looked at wage disparities, occupational segregation, and algorithmic discrimination. Galán for instance, conducted a qualitative review of gender-based violence in the future of work and found workplace harassment widespread but rarely examined in platform-specific contexts (Galán, 2024). Fairwork’s annual report mapped labor standards globally and concluded that safety provisions for women remain inadequately addressed (Fairwork, 2023). Nazareno reviewed the gendered impacts of ridesharing and flagged physical and psychological safety as understudied (Nazareno, 2025). None of these reviews, however, has pulled together the specific safety strategies and policies aimed at protecting female gig workers from sexual harassment, which is the gap this scoping review tries to fill. Studies do exist on gig worker safety, particularly in ride-hailing and food delivery, but none has focused specifically on female gig workers in platform-mediated work, and most of what does exist concerns physical safety tied to occupational hazards rather than harassment. There is a notable lack of research that addresses sexual harassment and sexual violence against female gig workers, which will be supported through a bibliometric analysis of the extracted evidence. Additionally, the literature is limited regarding policies implemented at the platform and state levels to ensure the safety of female gig workers at the workplace. Therefore, in the absence of legal safeguards, both at the state and aggregator levels, relating to the workplace safety of female gig workers, their work prospects are greatly undermined. This scoping review is pivotal as it attempts to build a narrative and tabular synthesis of existing evidence to underscore where the aggregators and state are lacking in addressing the safety of female gig workers from sexual harassment and sexual assault. This study would serve as a precursor for conducting a systematic literature review and meta-analysis in the future. Furthermore, this study is important because it suggests measures for the effective enactment and adoption of legal and policy frameworks to ensure the safety of female gig workers across platforms in the future.

Method

To evaluate the safety standards and measures adopted by workers, platforms, and the state to ensure the safety of female gig workers on app-based platforms, the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA-ScR) 2020 approach was adopted for this scoping review as depicted in Figure 1. All PRISMA-ScR (2020) guidelines were followed in this study. The database search was conducted on January 15, 2026, and the review covers literature published up to that date.

Inter-reviewer agreement was measured using Cohen's kappa: 0.82 at the title and abstract screening stage (strong agreement) and 0.78 at full-text screening (substantial agreement). Discrepancies went to a third reviewer for discussion, and a study was included only once at least two of three reviewers agreed on its eligibility. The data-extraction coding framework was built both inductively and deductively from the PRISMA-ScR data charting guidelines, with data coded under country/region, sector, study method, sample characteristics, safety concerns identified, site and perpetrator of risk, and safety strategies at the individual, collective, aggregator, and state levels. A study was included only if it described at least one strategy, practice, or policy response, however informal, adopted by any stakeholder to address female gig workers' safety concerns; studies that reported safety concerns without any corresponding response were excluded. Restricting the search to peer-reviewed journal articles, and excluding gray literature such as government policy documents, platform terms-of-service, NGO reports, and legal judgments, is a limitation of this review, made for the sake of methodological consistency and replicability. A separate policy-mapping exercise drawing on gray literature would be a useful next step for capturing regulatory and governance responses more fully. A narrative and tabular synthesis approach was used to describe the collected data. The authors' keyword co-occurrence network analysis, three-field plot, and thematic mapping using R 4.5.2 software were also conducted to underscore the scarcity of scholarship on the concerning issue of safety of female gig workers from sexual harassment at the workplace and sexual assaults. This highlights an urgent need to delve deeper into this area to ensure greater female workforce participation. The bibliometric analysis was performed using the bibliometrix package (version 4.1.4) in R software (version 4.5.2), which enables systematic mapping of scientific literature. The bibliometric dataset comprised all records retrieved from the three databases prior to deduplication ($n = 482$), exported in csv format. Data cleaning involved: (1) removing duplicate author keywords and standardizing variant spellings (e.g., "gig-economy" and "gig economy" were merged); (2) retaining only author-assigned keywords, excluding indexed keywords, to ensure analytical consistency; and (3) applying a minimum threshold of three co-occurrences for inclusion in the keyword co-occurrence network, following standard practice in bibliometric mapping. The co-occurrence network was normalized *via* the association strength index which allows the co-occurrence strength between different keywords to be meaningfully compared across different keywords by taking into consideration the volume of the keyword used. The three-field plot shows how the authors of publications are related to both the organizations from where the publications were published as well as the keywords that describe those publications (the minimum frequency of all three fields is two). The thematic map developed from Callon's centrality measures (the degree to which a theme interacts with other themes) and density measures (the degree to which a theme develops in a self-contained manner). The bibliometric results outlined in this paper help to substantiate the research gap identified in the review. There is a quantifiable gap in the literature pertaining to the intersection of gender, the gig economy and workplace safety from sexual harassment, which

correlates with the results in the traditional narrative synthesis discussed below.

Data sources

An advanced search of three databases, Scopus, Web of Science, and APA, was conducted using the same keywords ("gig economy" OR "app-based economy" OR "app-based work" OR "platform work" OR "gig work" OR "on-demand work" OR "digital platform work" OR "platform economy" OR "sharing economy" OR "on-demand economy" OR "digital labor" OR "digital freelancing" OR "ride hailing" OR "food delivery" OR "beauty services" OR "gig worker" OR "platform based worker" OR "app-based worker") AND ("women" OR "gender" OR "female" OR "woman" OR "females" OR "sex") AND ("safety" OR "security" OR "discrimination" OR "harassment" OR "violence" OR "assault" OR "protection" OR "policy" OR "governance") for title, abstract, and keyword searches, as presented in Table 1. The search string was formulated to capture the safety strategies and policies implemented by various stakeholders to address the issue of sexual harassment and sexual assault faced by female gig workers at workplace.

Inclusion and exclusion criteria

Studies that employed quantitative, qualitative, or mixed-method approaches were included if they met the following criteria (Table 2): (1) full-text articles, (2) articles published in English, (3) studies conducted until 2025, and (4) articles that focused on strategies addressing concerns about the safety and security of female gig workers from sexual harassment and assault in the workplace. Articles without full text, abstracts, books, book chapters, and review articles were excluded. Furthermore, studies that did not provide strategies to ensure the safety and security of female gig workers against sexual harassment and assault in the workplace were excluded.

Screening of data sources

The documents identified across various databases were consolidated into a single Excel file, and duplicates were manually removed by a reviewer. Subsequently, two reviewers assessed the titles and abstracts against the eligibility criteria. All extracted studies were independently reviewed by two reviewers. If both reviewers selected a study based on the inclusion criteria based on initial title and abstract screening, it advanced for full-text screening. Any disagreement pertaining to the inclusion of a study was resolved by discussion with a third reviewer after retrieval of the full text to determine its eligibility based on inclusion criteria. The dual review process ensured the retrieval of the most suitable and relevant articles for this study based on the pre-specified inclusion criteria. Following initial screening,

TABLE 1 Search strings.

Database	Search Strings for all Databases: title, abstract, and keywords. Filters: article; English	Number of records
Scopus	("gig economy" OR "app-based economy" OR "app-based work" OR "platform work" OR "gig work" OR "on-demand work" OR "digital platform work" OR "platform economy" OR "sharing economy" OR "on-demand economy" OR "digital labor" OR "digital freelancing" OR "ride hailing" OR "food delivery" OR "beauty services" OR "gig worker" OR "platform based worker" OR "app-based worker") AND ("women" OR "gender" OR "female" OR "woman" OR "females" OR "sex") AND ("safety" OR "security" OR "harassment" OR "violence" OR "assault" OR "protection" OR "policy" OR "governance")	326
Web of science		132
APA		24
Total		482

TABLE 2 Screening criteria.

Inclusion criteria	Exclusion criteria
Full-text peer-reviewed articles	Review articles, conference proceedings, books, book chapters
English language articles	Other than English language
Studies published up to year 2025	Studies conducted after 2025
Articles enumerating the safety and security strategies of female gig workers at workplace	Articles not expressly reporting safety and security strategies for female gig workers from sexual harassment and assault at workplace

two reviewers retrieved the full texts of the included studies for subsequent screening. Any conflicting stance regarding the relevance and suitability of the study for subsequent full-text screening was resolved by discussion with the third reviewer. Studies that examined any safety strategy implemented by gig workers either individually or collectively, as well as those employed by app-based platforms or the state, were included in this study. Disagreements arose at both the title/abstract and full-text screening stages and were of three main types. First, disagreements related to study eligibility occurred when a study focused on gig workers generally but did not explicitly address female workers or gender-specific safety concerns; such cases were resolved by consensus, with the third reviewer determining whether the study provided sufficiently relevant data on women's experiences. Second, disagreements concerning inclusion criteria arose for studies that examined occupational hazards or road safety without explicitly addressing sexual harassment or gender-based violence; these were excluded after third-reviewer discussion, as they did not meet the criterion of enumerating safety strategies specifically directed at protecting female gig workers from harassment or assault. Third, interpretation-related disagreements occurred when reviewers differed on whether a study "enumerated" a safety strategy or merely described a safety concern; the third reviewer adjudicated based on whether the study reported any deliberate action individual, collective, platform-based, or state-level taken to mitigate the identified risk. All disagreements were documented in a shared Excel sheet and resolved before advancing to the next screening stage.

Data items

Two reviewers jointly developed a table to chart the relevant items stated below to illustrate a comprehensive synthesis of the general descriptive details of the study and the safety and security strategies employed by various stakeholders to ensure the safety of female gig workers from sexual harassment, assault, and gender-based violence in the workplace.

The following details were extracted from the selected studies: author of the study, country in which the study was conducted, sector on which the study focused, method employed for conducting the study and data analysis, respondents' characteristics along with the focus area of the study, safety concerns of female gig workers, the site where they feel unsafe and from whom, and the safety strategies generally employed at distinct levels by various stakeholders in Tables 3 and 4 in data charting form developed jointly by the reviewers.

Results

Two reviewers jointly prepared the data charting form to document the safety concerns and strategies from the included studies. Both reviewers charted the data independently, and any conflict was resolved through discussion with a third reviewer. The present study reveals multiple safety concerns, namely abusive behavior in the workplace, restrictive safe working hours,

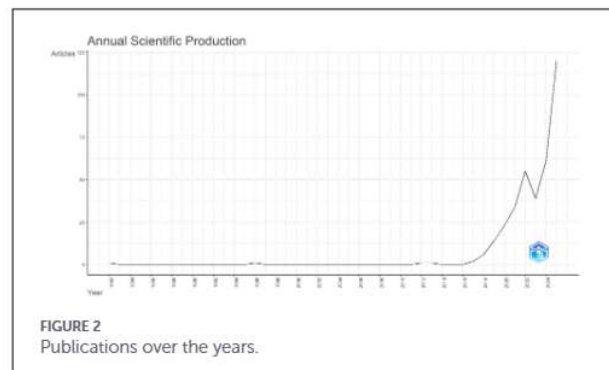
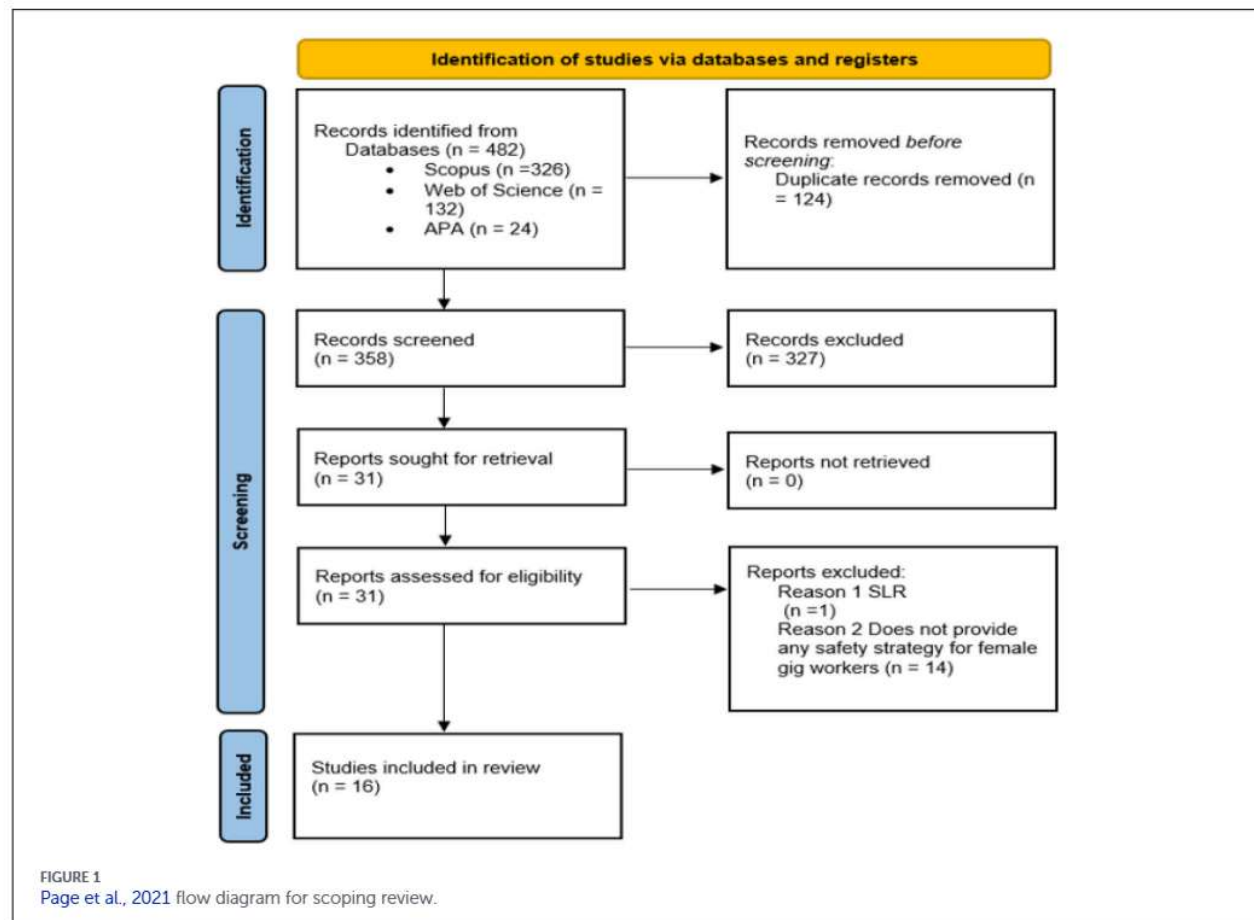
TABLE 3 Description of the included studies.

Author	Country	Sector	Method	Sample size	Respondents	Focus area
Zhuang et al. (2025)	Suzhou, China	Ride hailing	Accessing the local transportation bureau's database, nearly 12 billion trip records were collected, covering the entire October 2023. cell phone data were utilized to enhance spatial analysis; Spatial-temporal analysis; Statistical analysis of large datasets using Python	61,437 58,369(M) 3,104(F)	Ride-hailing drivers in Suzhou, China,	Perspective of Ride-hailing drivers in Suzhou, China
Liu and Huang (2025)	China	House cleaning	In-depth interviews; NVivo analysis	47(F)	Rural migrant women working as house cleaning workers	Perspective of women house cleaners working in Shanghai
Bakas and Salman (2024)	Türkiye	Tourism	In-depth interviews; thematic analysis proposed by Esfehani and Walters	10 4(M) 6(F)	Tourism based platform workers	Perspective of gig workers in tourism industry
Hiramatsu (2022)	Mexico	Ride-hailing, taxi services	Interview and focus group interviews; analysis using MAXQDA software	12 + 2 + 3(F)	"Female drivers state-certified taxis (7), tourist taxis (1), platform taxis (3) self-employed driver (1); two civil society organizations; staff of CENFES; leaders and staff of Laudrive and Ladyfy and a self-employed driver"	Perspective of female drivers
Beigi et al. (2020)	Tehran, Iran	Ride hailing	Semi-structured interviews; constant comparative method analysis using excel as NVivo cannot read Farsi language	34(F)	Female internet taxi drivers (FTD)	Perspective of Female internet taxi drivers (FTD)
Juddi et al. (2024)	Indonesia	Ride hailing	Semi structured interviews with 12 female drivers were conducted in Bandung, Indonesia; data was analyzed through a data interpretation process using data coding	12(F)	Female platform drivers	Perspective of female drivers
Putri et al. (2023)	Indonesia	Ride hailing	Structured interviews; thematic analysis	30	Drivers of three transportation related platforms	Perspective of drivers of three transportation related platforms
Martínez Caparrós (2024)	Spain	Ride-hailing	Ethnographic research based on Semi structured interview followed by narrative approach in data analysis	35(F)	Platform based cab drivers-15 Taxi drivers-20	Perspective of drivers

(Continued)

TABLE 3 (Continued)

Author	Country	Sector	Method	Sample size	Respondents	Focus area
Dhar and Thupplikkat (2022)	India	Beauty services	Qualitative research based on in depth interviews	11 1(M) 10(F)	Journalist who covered urban company protest 2021-1 Women trade unionist (AIGWU)-2 women workers-7 Male urban company worker-1	Perspective of women working for Urban company
Cordero (2022)	Ecuador	Food delivery	Both qualitative and quantitative methods were implied for data collection-considered oral history archive for collecting women delivery workers interview's which has already been done as well as survey method	348	on demand delivery workers	Perspective of demand delivery workers specific-focused mainly on women delivery workers issues n challenges
Anwar (2022)	Ghana, Kenya, Uganda, Nigeria, South Africa	Ride-hailing remote work- Upwork platform	Empirical research based on in depth interviews along with the implication of thematic analysis by using NVIVO software for coding and sub-coding	141 123(M) 18(F)	Ride hailing workers-76 Remote workers-65	Perspective of ride-hailing and remote workers.
Kwan (2022b)	China	Ride-hailing	Research is based on data collected through Chatnography in 17 months duration along with semi structured interviews conducted through social media and instant messaging platforms of Didi platform women drivers	30(F)	Ride-hailing drivers	Perspective of ride-hailing drivers
Stringhi (2022)	Global	Remote or web-based work	Ethnographic study by employing digital method approach for collecting statements mentioned on community discussion section of Upwork related to sexual harassment and gender-based violence		Women working on Upwork	Perspective of remote or web-based workers of Upwork
Hunt and Samman (2020)	South Africa	Domestic work	Qualitative study based on in depth interviews of domestic workers along with thematic analysis	36(F)	women domestic platform workers	Perspective of women domestic workers
Ajonbadi et al. (2025)	Nigeria	Location-based and Cloudwork platform work	Qualitative study based on interpretivist approach along with semi structured interviews	37 23(M) 14(F)	21 location-based 13 cloud work 3 both	Perspective of three location-based and four Cloudwork platform workers
Cai et al. (2024)	China	Ride-hailing	Systematic search using keyword and social media content databases; content analysis		Database retrieved studies and social media content	Feminist perspective



vulnerable job security, customer anonymity, and lack of client verification, which build an unsafe working environment for women engaged in the gig economy, as depicted in Table 3. In addition, this study explores various existing strategies employed by female gig workers at the individual and collective levels, along with the actions taken by aggregators to mitigate these safety concerns, as depicted in Table 4. Figure 2 presents the publication of literature on the gig economy over the years. It can be observed that after COVID-19, studies on the gig economy have increased in number and have gained momentum and increased exponentially.

Keyword co-occurrence network analysis

The keyword co-occurrence analysis examines the frequency with which two or more terms appear together in the same publication, thereby reflecting ongoing and contemporary research trends in the field. Figure 3 presents a keyword co-occurrence network map generated using R 4.5.2. The node limit was set to 30 to map the co-occurrence network of the author keywords. The nodes gig economy and gender are central themes because they are used most frequently by the authors. Additionally, these two nodes represent the strength of their co-occurrence in scholarship across the globe. Furthermore, the gender node has the highest link strength, representing its relationship with keywords such as gig economy, platform economy, sharing economy, future work, employment, and safety. The safety node is very small, suggesting the need for more studies connecting the keywords gig economy, gender, and safety.

Three-field plot

A three-field plot analysis was conducted using R 4.5.2 software to map the most relevant keywords, sources, and countries of authors to identify the most credible, influential, and critical contributions in the field of the gig economy (Figure 4). It also

TABLE 4 Safety concerns and strategies employed by various stakeholders.

Author	Safety concerns		Safety strategies at various level				
	Safety concern (site, inducer)		Workers: Individual	Workers: Collective	Aggregator	Government	Organizations/ Unions
Zhuang et al. (2025)	Industrial area during night time	Safety during night time	Shorter trips closer to home				
Liu and Huang (2025)	Home/place of work; older male; present alone	Sexual harassment- "sexual gaze," "being lightly touched," "being forcefully groped."	"Minimize eye contact" with men clients; "keep a physical distance"; "making big noise with cleaning tools"	"Earmarking the sex-crazed pigs"; "sharing the notes" about risk management; through WeChat			
Bakas and Salman (2024)	At work, pub crawl	Sexual and verbal harassment, phone lines do not respond due to time zone difference; lack of dedicated harassment reporting channels	Enlisting help from male during night; take friends along for pub crawl		Can only verify guest ID		
Hiramatsu (2022)	During ride, at parking stations; male clients; co-workers, locals	Sexual harassment and aggression, "They curiously stay looking at you, their eyes almost want to undress you,"; treated as sexual objects; "whether they are married," "take me to your home," or "whether they offer any other service besides taxi service."	Switch workplaces to another taxi stand or hotel				Civil society organization PROEM and MOM; CENFES provides training to improve female drivers' security and protection from risk

(Continued)

TABLE 4 (Continued)

Author	Safety concern (site, inducer)	Safety concerns				Safety strategies at various level			
		Workers: Individual		Workers: Collective		Aggregator		Government	Organizations/ Unions
Beigi et al. (2020)	During ride; suburbs; nearby cities; via text on mobile, male clients; passengers who looked suspicious	Risk of victimization and harassment—of possibly being sexually harassed and physically attacked while working alone and sharing a small space with strangers; male passengers gazed at their body	Preferred female passengers; putting a big hand bag on the front seat to avoid male passenger from sitting in front; boyfriend who was constantly alert and in touch with her by phone to make sure she was safe; carrying defensive tools; changed their car's front windows to smoke colors that did not allow outsiders to see inside the car; rejected passengers who seemed suspicious; "My car's doors are always locked; often don't work after 7 [pm] or 8 [pm]"			Initially, the internet-based taxi company assigned only female passengers to female drivers; later the company allowed female drivers to receive requests from both female and male passengers with the option of rejecting male passengers without being penalized for it. blocked the customers out of their system on reporting SH			
Juddi et al. (2024)	Customers, during ride	Susceptible to threats, assault, violence and harassment, no dedicated support service for female drivers	Pause operations, seek solidarity through group chats	Share problems and challenges on group chats		Grab collaborated with the National Women's Commission to combat sexual violence against female drivers and introduced a women-only fleet, "LadyGrab," for GrabExpress and GrabFood services in the Greater Jakarta area			

(Continued)

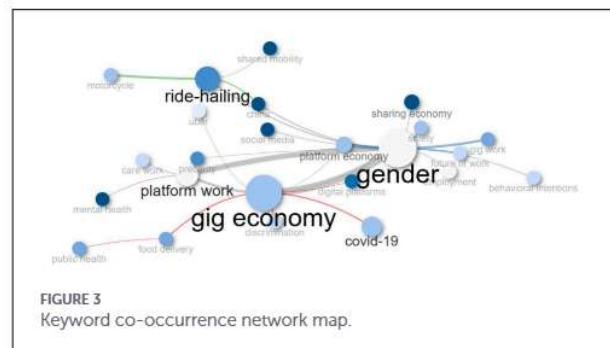
TABLE 4 (Continued)

Author	Safety concern (site, Inducer)	Safety concerns		Safety strategies at various level		
		Workers: Individual	Workers: Collective	Aggregator	Government	Organizations/ Unions
Putri et al. (2023)	Male customers	Female drivers were prone to harassment from male customers, but rejecting their advances could lead to a low rating as an act of retaliation.	Srikandi (a female-only group) to voice their concern has company representative	To address the challenge of inappropriate behavior by clients, Grab in partnership with The National Commission on Violence Against Women has offered women drivers training in the prevention and management of sexual violence and harassment as part of "LadyGrab" pro gram		
Martínez Caparrós, 2024	Cab; male passenger	Sexual assaults- involuntary touch unwanted flirting sexual objectification-monetary offerings for sexual services	Creating mental maps where to go where not to go- to avoid potentially dangerous routes; avoiding working at nighttime- trading income for their personal safety carrying anti rape spray; preparing positive mindset that nothing will go wrong before driving; becoming less feminine - adapting masculine behavior and appearance; texting their colleagues and sharing their location	Women only WhatsApp groups	Panic buttons	
Cordero (2022)	Public roads	Sexual Harassment; "they see you as Venezuelan and they already think you are willing to do anything" eroticisation				National strikes by Glovers_Ecuador
Dhar and Thupplikkat (2022)	Clients- home	Lack of client verification/anonymous booking risk	Double checking the client's details along with sharing their work locations with family members	WhatsApp groups; strike together to show their resistance while taking support of already existing workers union (AIGWU AND IFAT)	Emergency helpline number	Support by AIGWU and IFAT during strike by female workers

(Continued)

TABLE 4 (Continued)

Author	Safety concern (site, inducer)	Safety concerns				Safety strategies at various level			
		Workers: Individual		Workers: Collective		Aggregator		Government	Organizations/ Unions
Anwar (2022)	Roads, passengers	Carjacking risk at night, feeling of fear of being trapped in unsafe situation during night hours	Avoiding working during night hours, avoiding to accept bookings for certain dangerous route	WhatsApp groups, virtual meetings through zoom					Conducted international strikes with the solidarity of delivery workers from different countries
Kwan (2022b)	Male colleagues male and female passengers	Faced sexual exploitation through sexting on Common WeChat groups where both male and females are present, visiting remote areas at odd hours assault by female passenger, eroticization	Avoiding remote location rides to minimize the insecurity and danger	Women Only WeChat Group					Hong Kong Women Workers Association (HKKWA)
Stringhi (2022)	Client; Virtual workplace	Cyber stalking and harassment in the form of sextortion and the non-consensual dissemination of intimate imagery. Payment in return for performing sexual acts.	Blocking that user,						
Hunt and Samman (2020)	Client; Clients home, in transit	Physical and verbal use at client's home and threat on women worker's safety while coming back late at night from their workplace	Double checking the client's details along with sharing their work locations with family members avoid working late at night accept booking from those who are their known	WhatsApp group for regular updates					SADSAW (South African Domestic services and allied workers union) formed
Ajonbadi et al. (2025)		Ride hailing is considered unsafe for women by society		Social media groups Online community forum					AUATON- Amalgamated Union of App-Based Transporters of Nigeria
Cai et al. (2024)							Technology-enabled surveillance; record audio-video of entire trips		



provides insight into the sources that are actively supporting research on the gig economy and its stakeholders. Central themes, such as ride-hailing, gig economy, food delivery, and gender, have emerged as areas of interest for authors from Australia, China, the USA, the UK, and India.

Thematic map

A thematic map was created using R 4.5.2 to identify which themes are most relevant and have been developed with high centrality and which themes have been developed marginally. Figure 5 depicts that themes related to the gig economy, algorithmic management, and labor have been developed much more than gender inequality and discrimination. Themes such as ride-hailing, food-delivery, public health and safety (occupational) have high density and are considered motor themes that are most frequently worked upon. However, gender, gig economy, and safety in terms of sexual harassment at the workplace together do not form the core area of study; therefore, it needs to be explored in depth to address the issue of women's safety in the gig economy.

A bibliometric analysis of the databases clearly reveals a significant research gap and limited literature on the protection of female gig workers from sexual harassment in the workplace. Minimal research has been conducted on ensuring the safety of female gig workers. This observation aligns with our inclusion of only 16 studies that address the safety concerns of female gig workers and have examined the strategies employed by different stakeholders to protect them from workplace sexual harassment and sexual assault. This serves as a precursor for future research on the subject, as women constitute half of the world's population and can greatly contribute to the economic growth of a nation.

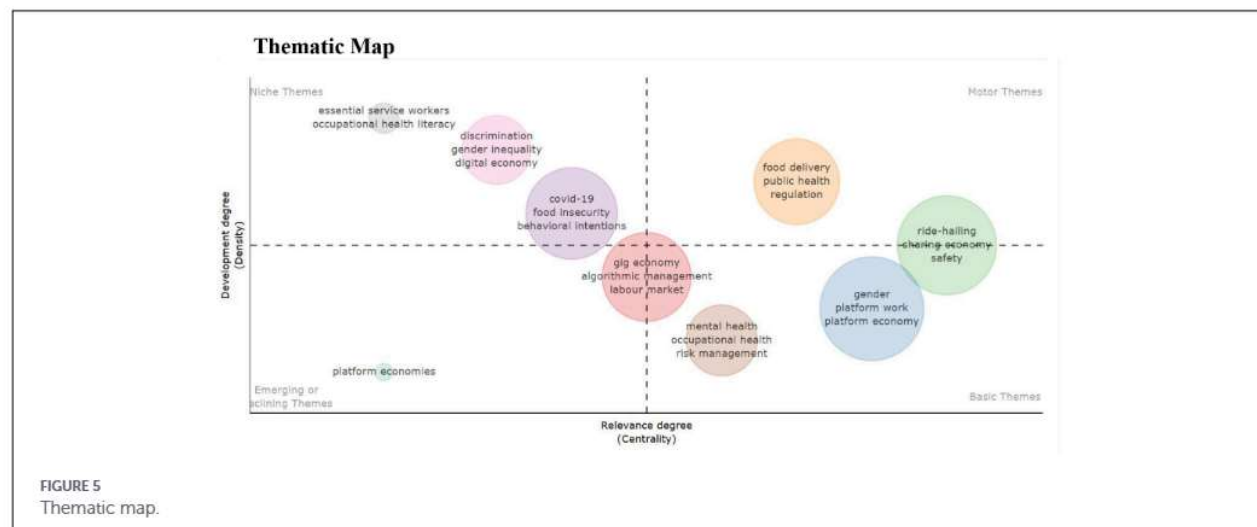
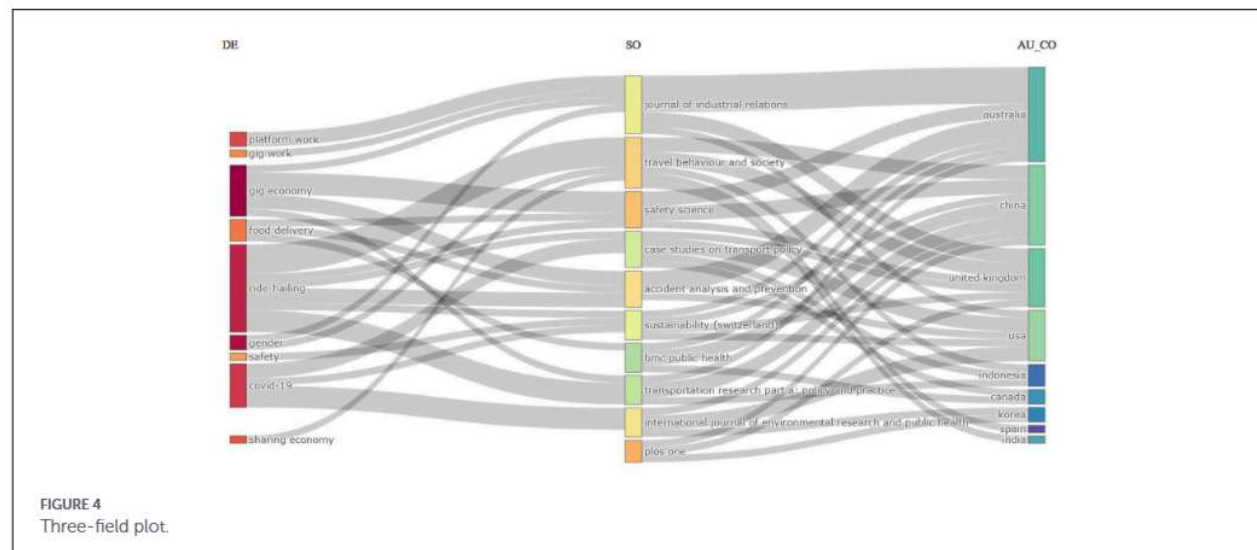
Workplace safety risks of women gig workers

Safety is one of the biggest barriers keeping women out of the gig economy (Nazareno, 2025). Women often participate in gig work because of the financial necessity and flexibility it offers to balance caregiving responsibilities, motivating them to

persist despite facing numerous safety risks and other substantial constraints (Juddi et al., 2024; Zhuang et al., 2025). They frequently navigate these challenges by ignoring them or refraining from lodging complaints about them. The safety concerns of female gig workers have been thematically analyzed under various headings, as reported below.

Abusive behavior

Women engaged in location-based services reported that their safety was challenged because of clients' abusive behavior in the workplace. Common safety risks encountered in their daily work routines included involuntary touching, gazing, verbal abuse, groping, unwanted sexual advances, sexual objectification, and propositions for sexual activities in exchange for money. Such gender-based violence exacerbates oppression and vulnerability, paralleling the experiences of women working in ride-hailing, on-demand delivery, and home services (Ajonbadi et al., 2025; Anjali Anwar et al., 2021; Cordero, 2022; Cordero and Daza, 2021; Hunt and Samman, 2020; Kwan, 2022b; Liu and Huang, 2025; Louzado-Feliciano et al., 2022; Martínez Caparrós, 2024; Nazareno, 2025; Wiesböck et al., 2023). In most instances, male clients are identified as the primary perpetrators of these activities, whereas female clients are generally involved in verbal abuse toward female workers (Juddi et al., 2024; Liu and Huang, 2025). Women employed in domestic services specifically note that their workplace insecurity often arises from interactions with male clients or the presence of male family members, with women sometimes making inappropriate comments or requesting additional services without compensation (Ajonbadi et al., 2025). Underlying much of this is patriarchal oppression, which keeps women excluded or subordinated in public institutions, including the workplace. These socio-cultural and political norms make workspaces feel masculine by default, creating discrimination and a range of informal and formal barriers for women in the gig economy (Anwar, 2022; Cordero, 2022). This sexist behavior is particularly evident in cab-hailing and on-demand delivery services, where patriarchal control enables male drivers to sexually harass female drivers through verbal abuse, vulgar language, infiltrating women-only social media groups by masquerading themselves as females, and sexting by obtaining their personal mobile numbers from common WhatsApp and WeChat groups (instant messaging applications), thereby marginalizing and socially excluding them from this profession (Hiramatsu, 2022; Kwan, 2022b). According to a 2021 survey, 66% of women in Africa engaged in online delivery work reported experiencing sexual harassment and numerous insults from male co-workers on public roads (Cordero, 2022). Similarly, women involved in ride-hailing services face continuous threats, verbal violence, and personal insults, which often lead to social media outrage (Cai et al., 2024; Tang et al., 2021; Yoo et al., 2024). This type of abusive behavior from clients is also prevalent among women who engage in remote work settings. In remote work settings, women frequently report various forms of cyber violence, including sexual harassment through online platforms, such as sextortion and the non-consensual dissemination of intimate imagery *via* personal phone numbers, as well as obscene comments



and flirting online (Stringhi, 2022). The exploitation of workers on digital labor platforms varies between the Global North and Global South because of the presence of more effective redressal mechanisms and labor rights. However, in the Global South, workers often lack a dependent working relationship with these platforms (Cordero, 2022).

Restrictive “safe” working hours

Women engaged in digital labor platforms often prefer working during daylight hours, a pattern that mirrors trends observed in the traditional employment sector. This inclination primarily stems from safety concerns associated with operating in these spaces. Such prioritization incurs certain costs, frequently resulting in increased earnings disparities between female gig workers and their male counterparts, who are more comfortable working late.

Consequently, the night becomes a “masculine time”, effectively excluding many women from urban workspaces (Beigi et al., 2020; Martínez Caparrós, 2024; Zhuang et al., 2025). Women often trade off potential income to ensure their personal safety by limiting their work to evening hours. The impact of these safety-driven choices extends beyond financial implications, affecting the physical and psychological wellbeing of female platform workers. Women working at night often encounter challenging situations, including interactions with abusive or intoxicated passengers, and some have faced carjacking situations (Anwar, 2022). These experiences alleviate health concerns and potential trauma, highlighting the complex interplay between economic necessity, personal safety, and occupational hazards for women engaged on digital platforms. Furthermore, studies have indicated that nighttime operations amplify fears of remote locations and unfamiliar places (Anwar, 2022). Consequently, women often restrict their services to certain areas and avoid traveling to the suburbs; these self-imposed restrictions on service areas may significantly impact women’s

earning potential, creating a cycle in which safety concerns directly affect the economic stability of women (Beigi et al., 2020; Kwan, 2022a).

“Brushing off” harassment *vis-à-vis* vulnerable job security

Digital platforms rely on algorithms that operate autonomously without direct human intervention. These algorithms are complex sets of instructions designed to process vast amounts of data and make decisions based on user behavior, preferences, and patterns. However, the autonomous nature of these algorithms and platforms has raised concerns regarding the safety and wellbeing of workers, particularly women (Bakas and Salman, 2024). Many women engaged with digital labor platforms have reported that the platform’s operational mechanism sometimes exacerbates threats to their safety. A key aspect of this is the use of rating and reputation scoring mechanisms to exert control over workers. These systems operate by exerting control over workers through rating and reputational scoring systems (Anwar, 2022; Kwan, 2022a; Mondon-Navazo and Murgia, 2025). Owing to the fear of poor ratings, work pressure, and job loss, women working in gig roles often have to tolerate clients’ abusive behavior and the risk of accidents, which frequently increases their levels of job dissatisfaction, anxiety, and other psychological risks (Laskaris et al., 2024). This situation also reflects the imbalanced power dynamics between clients and workers. In several studies, female gig workers mentioned that platforms favor clients over workers, and even a single bad comment and poor rating from a client can result in job loss (Anjali Anwar et al., 2021; Laskaris et al., 2024; Mondon-Navazo and Murgia, 2025). This power imbalance between clients and workers can lead to the development of a vicious cycle of exploitation and vulnerability for female gig workers, who must prioritize client satisfaction over their own wellbeing because of the constant pressure to maintain high ratings and positive feedback. Additionally, there is a lack of adequate redressal mechanisms and safety protocols provided by the platform company, which not only increases safety risks for women gig workers but also fosters a culture of impunity for clients who may engage in exploitative or abusive behavior (Dhar and Thuppilikkat, 2022). This undermines the overall integrity of the gig economy, hinders the potential to create a more equitable and secure work environment, and highlights the urgent need for reforms to create a more equitable and secure work environment.

Customers’ anonymity and lack of clients’ verification

The gig economy operates on a model that connects service providers with consumers directly through a common platform, thereby limiting face-to-face interactions between service providers and customers. This reduction in direct interaction can lead to

concerns regarding trust and accountability issues. The anonymity and lack of client verification inherent in this model pose significant security risks, particularly for women using location-based services. These individuals may inadvertently accept jobs from individuals with malicious intent, potentially exposing themselves to dangerous situations, which is further exacerbated by the solitary nature of many gig jobs, which can leave workers feeling vulnerable and exposed to potential risks (Hiramatsu, 2022). Women working in domestic services have reported feeling insecure, unsafe, and helpless due to isolated working conditions when entering strangers’ homes (Dhar and Thuppilikkat, 2022). Similarly, in the US and Mexico City, women involved in cab-hailing services have reported their experiences of assaults and sexual harassment attributed to the absence of a robust customer verification system (Hiramatsu, 2022; Louzado-Feliciano et al., 2022). Additionally, in Turkey, women who worked as platform-based tourist guides indicated that they lacked the autonomy to select their guests and received only ID cards and photographs of guests. Consequently, they are often assigned to manage large groups of male guests, which can be uncomfortable and unsafe, particularly when these guests request visits to pubs (Bakas and Salman, 2024). These circumstances reflect the existence of information asymmetry between the client and service provider, which alleviates safety risks for both parties (Wiesböck et al., 2023).

Key outcomes of safety strategies for female gig workers

Safety strategies comprise numerous practices that often act as a form of agency and resistance to the issues faced by female gig workers (Mondon-Navazo and Murgia, 2025). These preventive measures enable women to overcome potentially unsafe situations that are inherent in the gig economy while demonstrating their resilience and will to equal participation with men in gig workspaces. A careful examination of the studies revealed that there are several barriers that either obstruct the participation of female gig workers or are used as temporary engagements to supplement family income. Women have not been able to fully realize the employment potential of the platform economy because of the absence of gender-specific needs in terms of legislation, infrastructure, or grievance redressal, with special emphasis to safety-related provisions. Extensive research has shed light on various strategies designed to ensure the safety of women participating in the gig economy. These strategies function at three distinct levels: the individual level (gig workers), the aggregator level (platforms), and the collective level (women gig workers collectively).

Safety strategies at individual level

Women engaged in gig economy often practice numerous unique strategies at the individual level to ensure their safety. These defensive strategies vary according to different workspaces,

but are mainly practiced by women, particularly those engaged in location-based services (Anwar, 2022; Beigi et al., 2020). Various studies have identified that avoiding nighttime operations, remote locations, and restricting themselves to certain locations is the most common strategy. In addition to limiting working hours, other common strategies include sharing locations with friends and family, carrying defensive tools such as anti-rape sprays, and making noise to alert people in the surroundings (Anwar, 2022; Liu and Huang, 2025; Martínez Caparrós, 2024; Zhuang et al., 2025). Furthermore, no significant variation in these strategies has been identified between the Global North and South. For instance, women drivers in Kenya and Ghana have reported declining requests from certain localities of Nairobi and Accra, sometimes resulting in job and income loss due to account suspension (Anwar, 2022). Similarly, in China, women in the ride-sharing industry avoid driving at distant locations and during late hours to fulfill domestic and childcare commitments while minimizing feelings of insecurity and danger. Instead, they prefer to work during morning peak hours from 7 to 10 am, after dropping children at school, to maintain a high service rating (Kwan, 2022b). They often create mental maps, which are not geographical mappings but plans that help them decide where to go, where not to go, what to wear, and how to feel safer (Martínez Caparrós, 2024). This reflects that women often bear dual burdens and must trade off income to ensure their personal safety by restricting their operations to evening hours while balancing household and childcare responsibilities. Furthermore, another unique strategy involves adopting a “less feminine attitude.” Women engaged in ride-hailing services have mentioned that to fit into a masculine work environment, they must move away from traditional feminine behaviors, modify their appearance, and adopt traits associated with masculinity, such as language use, aggressiveness, toughness, thick skin, short hair, and dressing like men. By adopting these approaches, women drivers aim to maintain their personal safety and reduce the likelihood of workplace harassment (Beigi et al., 2020; Hiramatsu, 2022; Martínez Caparrós, 2024). In addition to the aforementioned strategies, there are other sets of specific strategies at the individual level employed by women working in the gig sector to ensure their safety. These include carrying defensive tools such as anti-rape spray, placing handbags on the front seat to avoid close proximity with male passengers, changing the car's front window to a smoke color, sharing their locations with friends and family, and being accompanied by male friends (Beigi et al., 2020; Martínez Caparrós, 2024; Mondon-Navazo and Murgia, 2025). Some studies have also shown that women often cope with the fear of victimization by deliberately avoiding thoughts of dangerous situations, thereby establishing mental preparedness as part of their daily routine before commencing work (Beigi et al., 2020; Liu and Huang, 2025). These defensive strategies are not completely new in nature, this approach exists in traditional workspaces, where women adopt similar strategies to build safer working environment for themselves (Moser et al., 2021; Turner et al., 2021). This clearly reflects that women are unsafe in both digital and traditional workspaces and are solely responsible for ensuring their own safety, which often increases their workplace challenges along with their mental health. These strategies often provide a sense of agency but also raise concerns about addressing the root causes of safety

issues that limit women's participation in traditional and platform-based workspaces.

Safety strategies at the aggregator level

Platform companies play a significant role in ensuring worker safety. They generally adopt technology-oriented measures to create a safer work environment for both passengers and workers in platform economy. Various studies have shown that respondents have marked the imbalance of power relations between workers and clients and the company's preference for a client's safety over the workers, which results in limiting the number of initiatives taken regarding the safety of women workers (Anjali Anwar et al., 2021; Mondon-Navazo and Murgia, 2025). Most of these initiatives are undertaken by platform companies that deal with location-based services because of the higher risk and potential threat to women's safety. This includes SOS helpline numbers, GPS installation for tracking, and installation of panic buttons along with automatic voice recording (Anjali Anwar et al., 2021; Beigi et al., 2020; Martínez Caparrós, 2024; Putri et al., 2023; Tandon and Sekharan, 2022). In China, *Didi*, a ride-hailing company, has launched an audio-video surveillance system to record entire trips to ensure passenger's safety, however this technology-enabled surveillance has the potential to cater the safety needs of women drivers also (Cai et al., 2024). Similarly, women engaged in platform-based home services have mentioned that, owing to platform surveillance, they have gained the trust of their family members to join this profession (Anjali Anwar et al., 2021). This also indicates that the platform surveillance approach has the potential to build a safer working environment and promote women's work participation, although very few companies have adopted this approach. In addition, a few ride-hailing and on-demand delivery companies have employed gender-segregated approaches to workspaces to ensure the safety of women working with them. This involves the segregation of services based on gender, offering separate fleets and services exclusively for women by transitioning from a traditional pooling system that matches riders with drivers without considering gender to a “hybrid system” that includes a female-only option (Tang et al., 2021). In Indonesia, Gojek introduced ladyGrab in collaboration with the National Women's Commission, followed by *Didi*—a female-only service in China, Shecabs in India, Shekab in Pakistan, and chariot for women services in the USA (Juddi et al., 2024; Kwan, 2022b; Putri et al., 2023; Tang et al., 2021). However, these measures vary according to the nature of gig work. Platforms dealing with remote work often implement digital safety measures such as secure communication channels, identity verification, and data encryption to protect online gig workers. For instance, the Upwork platform has introduced a flagging mechanism to ensure the safety of female online workers. Women can flag client profiles they perceive as potential threats to their safety, and these profiles are permanently blocked by the company (Beigi et al., 2020; Stringhi, 2022). A commonality can be observed in the initiatives of spreading awareness, which is achieved through organizing campaigns, educational and training programs related to safe riding, dealing with traffic accidents, and anti-violence

against women by location-based service provider companies and remote work platforms (Anjali Anwar et al., 2021; Juddi et al., 2024; Shade, 2018). Although these initiatives serve as preventive measures, there is a noticeable gap in the development of effective reporting mechanisms for incidents that occur despite them being in place. Notably, in Nigeria, a platform-based company created a separate application where workers could register complaints and report incidents, followed by an SOS line provided by the Urban Company in Delhi (Ajonbadi et al., 2025; Tandon and Sekharan, 2022). Without a reliable reporting system, responses to safety concerns are likely to be slow or absent altogether.

Safety strategies at a collective level

Women engaged in the gig economy have developed innovative and distinctive strategies to navigate the challenges inherent to platform-based workspaces. These strategies represent proactive measures that women generally adopt in the absence of a formal redressal mechanism to address issues such as sexual harassment, thereby reflecting their resilience and adaptability in an environment that often lacks a structured support system. By leveraging communication technologies, women have created their own informal networks and channels to share information, provide mutual support, and foster safer working conditions (Ajonbadi et al., 2025). In China, women are using WeChat groups along with Sina Weibo, a social media platform, to resist gender-based violence (Kwan, 2022b). The practice of utilizing social media platforms and forming informal groups is common among countries in the Global North and South. Platform-based home cleaners in Berlin have named their group *Sindicat Helping*, followed by *Srikandi*, an informal WhatsApp group created by Gojek women ride-hailers (Cai et al., 2024; Putri et al., 2023; Mondon-Navazo and Murgia, 2025). These common communicative spaces often boost solidarity and mutual support among women gig workers and enable them to raise their voices collectively against misconduct (Cai et al., 2024; Jing and Susanne Yuk-Ping, 2025; Kwan, 2022a; Martínez Caparrós, 2024; Tandon and Sekharan, 2022). Furthermore, the utilization of social media as a collective safety strategy enables women workers to reach a larger audience and facilitates large-scale connectivity across borders (Cai et al., 2024; Cordero, 2022). Additionally, regarding the informal voice channels that women have developed using communication technologies, some studies have mentioned that women have also created formal voice channels by creating their own workers' unions that generally raise their concerns and provide services that help them balance their household responsibilities. For instance, the Hong Kong Women Workers' Association provides crèche facilities for women to work long hours. These practices indicate that women are opting for pragmatic activist approaches to raise their concerns and gain the attention of the government and relevant authorities (Ajonbadi et al., 2025; Cai et al., 2024; Hiramatsu, 2022; Hunt and Samman, 2020; Kwan, 2022b). Unions are primarily responsible for raising concerns with management and mediating between workers and authorities, but no formal

structure specifically addresses sexual harassment in the gig economy, which is a real gap in worker-protection policy. Unions and advocacy groups need to push for policies that actually safeguard female platform workers from harassment and discrimination, rather than leaving the issue to informal solidarity alone.

Although the review report various safety concerns of gig workers and safety strategies employed by various stakeholders based on included studies, it suffers from limitations, such as the fact that the included studies were not critically appraised for their quality. Furthermore, since the objective of the study was to include only those studies that examined any safety or security measures implemented by different stakeholders involved in the gig economy, only 16 studies met the inclusion criteria, thus excluding most studies. In addition, no prior protocol is mandatory for conducting a scoping review, and the review is up to date until 2025. A further limitation relates to the reference base. While the included studies span multiple regions and gig sectors, the manuscript's analysis would benefit from stronger engagement with the following bodies of scholarship: (1) platform governance and algorithmic accountability literature, which provides theoretical grounding for understanding how platform architectures produce and sustain gendered power asymmetries; (2) labor precarity and informal worker protection scholarship, which contextualizes the absence of formal safety structures within the broader political economy of gig work; and (3) gender-based violence at work literature, which documents the psychological and behavioral dimensions of harassment in non-traditional workplaces. It is also worth noting that risks differ substantially across gig work types: sexual harassment in location-based, in-person services (ride-hailing, domestic work, and food delivery) is a qualitatively different problem from cyber harassment on online and cloudwork platforms (Upwork, remote freelancing). Since ride-hailing studies make up most of the included evidence, their findings should not simply be assumed to hold for cloudwork or web-based gig contexts, and future reviews and policy work should treat this sectoral split explicitly.

Conclusion and way forward

This study set out to identify the safety concerns of female gig workers, and several stand out. Abusive behavior in the workplace is a direct threat to women's dignity and wellbeing. Fear of working at night reflects how much more vulnerable women feel during extended hours, and platforms make this worse by prioritizing efficiency over worker safety. Road safety matters a great deal for those in delivery and transportation work. Risk for women working in the gig economy increases due to the anonymity of customers and poor authentication processes for clients. As a result, some women may find themselves in vulnerable situations where they cannot validate a client's identity. This ultimately creates significant obstacles for women wanting to work in the gig economy, impacting their income and career progression. Additionally, the research also examined the safety measures used by workers, aggregators and governments,

and found that the combined result requires a multi-pronged approach; whereby, workers develop their own safety practices, aggregators improve the security of the individual platforms they operate, and governments implement suitable policies and regulations. The intent of identifying these strategies is to find initiatives that will allow for women to have increased safety and equal opportunity in the gig economy. To substantiate the findings of this review, some of the distinctions made in the evidence provided are; The results presented in this report support conclusions drawn from the 16 studies included in this research project. The main findings related to gig workers are the following: (a) women (female gig workers), who provide services such as ride-hailing or domestic work (maid or cleaning services), beauty services, food delivery, or remote platform work, are subjected to sexual harassment, verbal abuse, cyber violence, and physical threats from male customers/clients or male coworkers; (b) women are likely to use individual-level coping strategies (like avoiding working night shifts, using location-sharing applications, carrying defensive items) thus carrying much of the personal responsibility for their own safety; (c) there are safety mechanisms in place at the aggregator level (e.g., SOS/panic buttons, GPS-based tracking systems, gender-segregated fleets, and flagging systems), but these mechanisms are largely reactive, sporadically implemented, and inadequate operationally in regards to the time to respond; and (d) WhatsApp, WeChat groups, and informal unions provide collective strategies to gig workers and meaningful solidarity with one another, but they lack formal institutional power. Authors inferred the following from available evidence and not specifically within the studies alongside the evidence: a structural gap in platform governance has formed (1) due to misclassification of gig workers into independent contractors limits platforms' liability for gig workers' misconduct, so they do not have to provide them with worker protections under law; and (2) that rating-based punishment systems enable and only normalize, but do not provide solutions for harms occurring through normalizing harassment; and (3) that there continue to be safety mechanisms (e.g., warnings, notifications of member activity) which do not provide institutionally meaningful safety measures, but are just a response to their societal expectations. Further, the following recommendations require further empirical investigation or legal analysis before implementation: (a) whether reclassifying gig workers as employees in specific jurisdictions would effectively extend sexual harassment protections without displacing workers from the platform economy; (b) the effectiveness of mandatory client verification and platform-audited grievance mechanisms as deterrents; and (c) the governance design of multi-stakeholder safety frameworks incorporating state, platform, worker union, and civil society representation. It should also be noted that risks differ meaningfully across location-based, online, and cloudwork platform types, and sector-specific policy responses may be needed for ride-hailing, domestic work, beauty services, and remote work contexts respectively.

From the included literature, it is clear that most strategies are informal and are utilized by female gig workers at the individual and collective levels. It can be further deduced that the strategies provided by a few aggregators, such as helpline numbers or panic

buttons, are inefficient with respect to the response time. A late response or no response further exacerbates the vulnerability of female gig workers. In this context, it is interesting to note that the solidarity approach among fellow gig workers is a more effective strategy for mitigating safety concerns than that provided by platforms. However, it may not always be feasible to be accompanied by a friend or family member, given the diverse nature of the workplace, which includes private households in addition to urban public places. In addition, co-workers may not always be free to address the safety concerns raised by female gig workers.

Urban public spaces have experienced greater expansion of platform services. As these spaces are part of the city, the safety of female gig workers imposes a liability on the state to protect and ensure the safety of its workers. The included studies clearly state that there is a lack of regulatory, formal, legal, and policy frameworks to address the safety and security challenges faced by female gig workers. The state must ensure their safety at both the law-and-order level and, more importantly, at the governance and planning levels. Various studies have pointed out that owing to the lack of accessible washrooms or rest areas in public spaces to cater to the feminine needs of female gig workers, their safety needs are at risk. It has also been pointed out that certain areas are perceived to be unsafe by female gig workers, especially ride-hailing, owing to the night or darkness caused by inappropriate lighting and the lack of CCTV provisions. In addition, women workers involved in beauty services or house cleaning services fear their safety because of the lack of safe transportation to and from the client's place. These issues and challenges are not law- and order-related but greatly concern the governance and planning of urban spaces and infrastructure development. To keep female gig workers safe in urban spaces, city planning needs to ensure streets are well lit and that visibility is not blocked by unplanned vegetation, unauthorized billboards, or other encroachments.

Although new labor legislation is being enacted or amended globally, the approaches vary significantly across regions. For instance, in India, the Code on Social Security (2020) emphasizes providing social protection to gig workers, whereas in Canada, the Digital Platform Workers Rights Act (2022) ensures minimum rights for these workers. In the United States, California Assembly Bill 5 (2019) illuminates the reclassification of gig workers as employees; however, Proposition 22 (2020) overrules Assembly Bill (2019) by excluding ride-hailing and delivery riders, thereby maintaining their classification as independent contractors. Within the European Union, the Platform Work Directive (2024) serves as a comprehensive framework that enlightens gig workers' employment status and introduces the rights of algorithmic management of gig workers. However, its implementation remains fragmented because of significant variations across member states. As exemplified by the Netherlands, judicial rulings have determined that Uber drivers fall under contractual labor contracts, despite the absence of unified comprehensive legislation, followed by Spain's Riders Law (2021), which targets workers' employment status by reclassifying them as employees. In contrast, France resisted complete reclassification, instead relying on case law and contractual models, despite the absence of unified comprehensive legislation. Collectively, these diverse legislative

measures highlight a critical gap: there is no express legislation dealing with the safety of female gig workers from sexual harassment at the workplace, which is otherwise available to female employees in traditional employer-employee relationships. This non-applicability of existing legislation is due to the dilemma surrounding the status of gig workers, whether they are employees, independent contractors, or delivery partners outside the purview of the traditional employer-employee relationship. It can be concluded that the state must either enact new legislation or amend its existing laws to encompass female gig workers within the ambit of traditional employer-employee relationships and provide them with the status of employees, given the level of control exercised by these platforms over the operations of gig workers, to grant them protection from sexual harassment at the workplace. It is imperative that platforms integrate built-in features on their apps to facilitate emergency assistance and prompt response delivery on site and ensure immediate and effective grievance redressal mechanisms that adequately address the representation of female gig workers. Additionally, the state, in collaboration with platforms, should develop inclusive infrastructure that caters to the gender-specific needs of women, such as sanitation facilities and resting areas. To ensure and enhance accessibility and safety across cities, particularly at night, city and municipal governments should install CCTV cameras and conduct regular patrols to ensure the safety and security of female gig workers. This scoping review sets the stage for future investigations into this topic. The range of legislative approaches to safety governance discussed indicates that legal reform alone is insufficient for effective safety governance. This indicates that if platforms (such as ridesharing companies) are to provide effective safety solutions, they must do so in the context of broader institutional frameworks. As such, the safety mechanisms offered by platforms (e.g., SOS buttons, helplines, and anti-harassment training) run the risk of becoming symbolic if they are not integrated into broader institutional frameworks (that include mandatory reporting, third party audits/evaluations, worker representation in platform governance and state enforcement; Putera et al., 2025). Consequently, in order for effective safety governance to occur, three elements must be present within a hybrid “multi-actor institutional” model—platforms must be held accountable through legally enforced minimum safety standards; workers’ unions and third sector organizations must be involved in the governance and grievance redressal systems, and government agencies must provide both regulation and development of gender-sensitive urban infrastructure. Such a model must be locally embedded recognizing that the risks facing female gig workers differ across ride-hailing in Mexico City, domestic work in South Africa, beauty services in India, and cloudwork in Nigeria rather than relying on a single universal technical solution.

Given that the gig economy is rapidly expanding and is serving as the most significant source of employment for women, it is imperative to encourage greater female participation as gig workers. The presence of female co-workers provides a sense of solidarity and safety for fellow female gig workers. Consequently, the state must re-evaluate its urban planning policy and enact or amend its existing legislation to ensure the safety of female gig workers at the workplace from sexual harassment in

both web-based and location-based services and encourage their workforce participation.

Policy and structural implications

A synthesis of the reviewed studies on safety strategies across individual, aggregator, and collective levels reveals several integrated outcomes that highlight existing deficiencies within Pulling together the safety strategies used at the individual, aggregator, and collective levels points to several gaps in the regulatory framework. At the individual level, the burden of staying safe falls disproportionately on women themselves, who adapt their behavior, avoid late-night shifts, or carry pepper spray thereby trading income for safety by limiting their working hours. Platforms, for their part, mostly react after the fact: helpline numbers, SOS buttons, and surveillance systems don’t do much to shift the underlying power imbalance between clients and workers. Collective strategies show real solidarity through digital groups and emerging unions, but they still lack any formal institutional backing. Put together, these gaps point to a structural problem in platform governance, where the interventions in place tend to create a sense of safety without addressing what actually makes the work unsafe.

The findings of this study are significant because they will serve as a reference point for policymakers and platforms to formulate policies to ensure the safety of female gig workers in the workplace, thereby ensuring their participation in the platform economy. The lack of safety governance for female gig workers according to both states and platform providers shows that collaborative and multi-stakeholder governance efforts are needed. To achieve effective, legitimate, sustainable, and institutionally embedded safety outcomes, policies based only on state laws or self-regulation of the platform (without active involvement of worker unions, civil society organizations, and communities affected by their use) will be of limited success (Valentina et al., 2025). The design of future female gig worker safety policies should therefore prioritize collaborative approaches rather than being based only on establishment of top-down regulations.

Data availability statement

The original contributions presented in the study are included in the article/Supplementary material, further inquiries can be directed to the corresponding author.

Author contributions

DB: Conceptualization, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing, Supervision. UG: Conceptualization, Supervision, Validation, Writing – original draft, Writing – review & editing. AM: Formal analysis, Methodology, Software, Writing – original

draft, Writing – review & editing, Visualization. KV: Formal analysis, Visualization, Writing – original draft, Writing – review & editing. VJ: Data curation, Writing – original draft, Writing – review & editing. PD: Data curation, Writing – original draft, Writing – review & editing.

Funding

The author(s) declared that financial support was not received for this work and/or its publication.

Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

The author(s) declared that Generative AI was not used in the creation of this manuscript.

Any alternative text (alt text) provided alongside figures in this article has been generated by Frontiers with the support of artificial intelligence and reasonable efforts have been made to ensure accuracy, including review by the authors wherever possible. If you identify any issues, please contact us.

Publisher's note

All claims expressed in this article are solely those of the authors and do not necessarily represent those of their affiliated organizations, or those of the publisher, the editors and the reviewers. Any product that may be evaluated in this article, or claim that may be made by its manufacturer, is not guaranteed or endorsed by the publisher.

Supplementary material

The Supplementary Material for this article can be found online at: <https://www.frontiersin.org/articles/10.3389/forgp.2026.1838133/full#supplementary-material>

References

- Ajaiyeoba, I. O. (2024). Diversity and emotional labor in the gig economy. *Equal. Divers. Inclusion*. 44, 777–791. doi: 10.1108/EDI-11-2023-0394
- Ajonbadi, H. A., Adekoya, O. D., Mordi, C., Cronk, H., Islam, M. A., Taofik, I., et al. (2025). Exploring the voice and representation mechanisms of platform workers and implications for decent work in the Nigerian gig economy. *J. Indust. Relat.* doi: 10.1177/00221856251337191
- Anjali Anwar, I., Pal, J., and Hui, J. (2021). Watched, but moving: platformization of beauty work and its gendered mechanisms of control. *Proc ACM Hum Comput Interact.* 4:250. doi: 10.1145/3432949
- Anwar, M. A. (2022). Platforms of inequality: gender dynamics of digital labour in Africa. *Gend. Dev.* 30, 747–764. doi: 10.1080/13552074.2022.2121059
- Assembly Bill (2019). California State Legislature (2019). Assembly Bill No. 5, Chapter 296: Worker status: employees and independent contractors (Cal. Lab. Code ? 2750.3). Approved by the Governor September 18, 2019.
- Ayentimi, D. T., Abadi, H. A., and Burgess, J. (2023). Decent gig work in Sub Sahara Africa? *J. Industr. Relat.* 65, 112–125. doi: 10.1177/0022185622111693
- Bakas, F., and Salman, D. (2024). 'You ensure your own safety': gender, fear and tourism labour in the gig economy in Türkiye. *Gender, Place Culture*. 31, 1072–1094. doi: 10.1080/0966369X.2024.2327999
- Banik, N., and Padalkar, M. (2021). The spread of gig economy: trends and effects. *Foresight STI Govern.* 15, 19–29. doi: 10.17323/2500-2597.2021.1.19.29
- Barzilay, A. R., and Ben-David, A. (2017). Platform inequality: gender in the gig economy. *SSRN Electron. J.* 47, 427–429. doi: 10.2139/ssrn.2995906
- Beigi, M., Nayyeri, S., and Shirmohammadi, M. (2020). Driving a career in Tehran: experiences of female internet taxi drivers. *J. Vocat. Behav.* 116:103347. doi: 10.1016/j.jvb.2019.103347
- Cai, Y., Xu, J. L., and Drinkall, S. (2024). Ride-hailing while female: negotiating China's digital public sphere. *Convergence*. 30, 1813–1831. doi: 10.1177/13548565231205976
- Churchill, B., and Craig, L. (2019). Gender in the gig economy: men and women using digital platforms to secure work in Australia. *J. Sociol.* 55, 741–761. doi: 10.1177/1440783319894060
- Collins, P. H. (2019). *Intersectionality as Critical Social Theory*. Durham, NC: Duke University Press. doi: 10.1515/9781478007098
- Cook, C., Diamond, R., Hall, J. V., List, J. A., and Oyer, P. (2021). The gender earnings gap in the gig economy: evidence from over a million rideshare drivers. *Rev. Econ. Stud.* 88, 2210–2238. doi: 10.1093/restud/rdaa081
- Cordero, K. H. (2022). Decolonial readings of platform economies: the organising of on-demand delivery women workers in Ecuador. *J. fur Entwicklungspolitik*. 38, 171–192. doi: 10.20446/JEP-2414-3197-38-2-171
- Cordero, K. H., and Daza, C. S. A. (2021). Migrant woman sustaining la lucha: a feminist takeover of the organization process of on-demand delivery workers in Ecuador. *J. Labor Soc.* 24, 579–597. doi: 10.1163/24714607-bja10013
- Crenshaw, K. (1989). *Demarginalizing the Intersection of Race and Sex: A Black Feminist Critique of Antidiscrimination Doctrine, Feminist Theory and Antiracist Politics Recommended Citation Crenshaw, Kimberle () "Demarginalizing the Intersection of Race and Sex: A Black Feminist Critique of Antidiscrimination Doctrine, Feminist Theory and Antiracist Politics. University of Chicago Legal Forum. Available online at: http://chicagounbound.uchicago.edu/uclfhhttp://chicagounbound.uchicago.edu/uclfh/vol1989/iss1/8 (Accessed January 18, 2026).*
- Cyrille, S., Assaf, G., Mathilde, P., and Rafael, V. (2019). *Gig Economy Platforms: Boon or Bane?*. Vol. 1550 (OECD Economics Department Working Papers). Available online at: https://www.oecd.org/en/publications/gig-economy-platforms-boon-or-bane_fdb0570b-en.html (Accessed January 18, 2026).
- Dhar, D., and Thuppilikkat, A. A. (2022). Gendered labour's positions of vulnerabilities in digital labour platforms and strategies of resistance: a case study of women workers' struggle in Urban Company, New Delhi. *Gend. Dev.* 30, 667–686. doi: 10.1080/13552074.2022.2127574
- Dokuka, S., Kapuza, A., Sverdlov, M., and Yalov, T. (2022). Women in gig economy work less in the evenings. *Sci. Rep.* 12:8502. doi: 10.1038/s41598-022-12558-x
- Dong, Z., Peng, J., and Liu, S. (2024). Gender pay gap in the gig economy. *China Finan. Econ. Rev.* 13, 3–22. doi: 10.1515/cfer-2024-0001
- Duggan, J., Sherman, U., Carbery, R., and McDonnell, A. (2020). Algorithmic management and app-work in the gig economy: a research agenda for employment relations and HRM. *Human Res. Manag. J.* 30, 114–132. doi: 10.1111/1748-8583.12258
- Eagly, A. H., Wood, W., and Diekmann, A. B. (2000). "Social role theory of sex differences and similarities: a current appraisal," in *The Developmental Social Psychology of Gender*, eds. T. Eckes, and H. M. Trautner (Mahwah, NJ: Erlbaum), 123–174.

- Fairwork. (2023). *State of the Global Platform Economy FairWork Annual Report 2023*. Berlin: Germany.
- Foong, E., Vincent, N., Hecht, B., and Gerber, E. M. (2018). Women (still) ask for less: gender differences in hourly rate in an online labor marketplace. *Proc. ACM Hum.-Comput. Interact.* 2:1–53. doi: 10.1145/3274322
- Galán, S. (2024). Gender-based violence in the context of the future of work: a qualitative review of the literature. *Trauma Violence Abuse*. 25, 4216–4229. doi: 10.1177/15248380241271374
- Galperin, H. (2021). “This gig is not for women”: gender stereotyping in online hiring. *Soc. Sci. Comput. Rev.* 39, 1089–1107. doi: 10.1177/0894439319895757
- Graham, M., and Shaw, J. (2017). *Towards a Fairer Gig Economy*. London: Meatspace Press. doi: 10.58704/5zv9-7s84
- Hau, M. F., and Borello, A. (2024). Between coping and resistance: migrant networks and alternative forms of collectivism. *Econ. Ind. Democr.* 45, 1067–1089. doi: 10.1177/0143831X231216001
- Hill, K. (2021). “Algorithmic Insecurity, Schedule Nonstandardness, and Gig Worker Wellbeing,” in *Population Association of America Poster Presentation*. Working Paper.
- Hiramatsu, A. (2022). Female taxi drivers in Mexico City: facing patriarchal structures as a force of oppression. *Gend. Work Organ.* 29, 1638–1657. doi: 10.1111/gwao.12633
- Hunt, A., and Samman, E. (2020). Domestic work and the GIG economy in South Africa: old wine in new bottles? *Anti-Traffick. Rev.* 15, 102–121. doi: 10.14197/atr.201220156
- Huws, U., Spencer, N. H., and Syrdal, D. S. (2018). Online, on call: the spread of digitally organised just-in-time working and its implications for standard employment models. *New Technol. Work Employ.* 33, 113–129. doi: 10.1111/ntwe.12111
- ILO. (2021). *World Employment and Social Outlook 2021: the role of digital labour platforms in transforming the world of work*. Available online at: https://www.ilo.org/sites/default/files/wcmsp5/groups/public/@dgreports/@dcomm/@publ/documents/publication/wcms_771749.pdf (Accessed January 16, 2026).
- Jing, W., and Susanne Yuk-Ping, C. (2025). Gendered labor contentions in the platform economy: studying food-delivery and house-cleaning workers. *Int. J. Comp. Sociol.* 66, 734–750. doi: 10.1177/00207152251351975
- Juddi, M. F., Maharani, A. J., Malunlana, M. A., and Pramiyanti, A. (2024). Communication patterns between companies and female drivers pre- and post-COVID-19 pandemic: evidence from Indonesia. *Howard J. Commun.* 32, 1–19. doi: 10.1080/10646175.2024.2368140
- Kwan, H. (2022a). Gendered precarious employment in China's gig economy: exploring women gig drivers' intersectional vulnerabilities and resistances. *Gend. Dev.* 30, 551–73. doi: 10.1080/13552074.2022.2118464
- Kwan, H. (2022b). Women's solidarity, communicative space, the gig economy's social reproduction and labour process: the case of female platform drivers in China. *Crit. Sociol. (Eugene)*. 48, 1221–1236. doi: 10.1177/08969205221101451
- Lain, B. (2024). Addressing the sharing economy—some (potential) inconsistencies of its emancipatory defense. *Philosophies*. 9:180. doi: 10.3390/philosophies9060180
- Laskaris, Z., Hussein, M., Stimpson, J. P., Vignola, E. F., Shahn, Z., Cohen, N., et al. (2024). A price too high: injury and assault among delivery gig workers in New York City. *J. Urban Health*. 101, 439–450. doi: 10.1007/s11524-024-00873-9
- Levine, S. M. (2025). Gig-economy myths and missteps. *Yale Law J.* 134:331.
- Lin, X., Lei, M., and Wang, X. (2023). How platform economic dependence leads to long working time: the role of work pressure and platform HRM practices. *Sustainability (Switzerland)*. 15:2634. doi: 10.3390/su151612634
- Liu, Y., and Huang, H. (2025). Emotional labor as a situated social practice: investigating the performance of emotional labor by women migrant workers in China's platform-based gig economy. *Gender Soc.* 39, 226–256. doi: 10.1177/08912432251317160
- Louzado-Feliciano, P., Santiago, K. M., Ogunsina, K., Kling, H. E., Murphy, L. A., Schaefer Solle, N., et al. (2022). Characterizing the health and safety concerns of U.S. Rideshare drivers: a qualitative pilot study. *Workplace Health Saf.* 70, 310–318. doi: 10.1177/21650799221076873
- Ma, N. F., Rivera, V. A., Yao, Z., and Yoon, D. (2022). “Brush it off”: how women workers manage and cope with bias and harassment in gender-agnostic gig platforms,” In *CHI conference on human factors in computing systems* (New York, NY: Association for Computing Machinery), 1–13. doi: 10.1145/3491102.3517524
- Manyika, J., Lund, S., Bughin, J., Robinson, K., Mischke, J., Mahajan, D., et al. (2016). *Independent Work: Choice, Necessity, and the Gig Economy*. Chicago, IL: McKinsey and Company.
- Mark, G., and Shaw, J. (2017). *Towards a Fairer Gig Economy*. Manchester, UK: Meatspace Press.
- Martínez Caparrós, B. (2024). Navigating the city: gendered work experiences in urban spaces. *Gend. Dev.* 32, 289–309. doi: 10.1080/13552074.2024.2348388
- Mondon-Navazo, M., and Murgia, A. (2025). ‘It's labour exploitation, but fortunately there is a lot of solidarity’. A qualitative case study with cleaning platform workers in Berlin. *Globalizations*. 1–19. doi: 10.1080/14747731.2025.2485788
- Moser, C. E., Branscombe, N. R., and Moser, C. (2021). Male allies at work: gender-equality supportive men reduce negative underrepresentation effects among women. *Soc. Psychol. Personal. Sci.* 13:372–381. doi: 10.1177/19485506211033748
- Muldoon, J., and Raekstad, P. (2023). Algorithmic domination in the gig economy. *European J. Polit. Theory*. 22, 587–607. doi: 10.1177/14748851221082078
- Nazareno, L. (2025). The gendered impacts of ridesharing in Brazil. *J. Indust. Relat.* 67, 661–686. doi: 10.1177/00221856241303769
- Newlands, G. (2024). “This isn't forever for me”: perceived employability and migrant gig work in Norway and Sweden. *Environ. Plan. A*. 56, 1262–1279. doi: 10.1177/0308518X221083021
- Page, M. J., McKenzie, J. E., Bossuyt, P. M., Boutron, I., Hoffmann, T. C., Mulrow, C. D., et al. (2021). The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ* 372:n71. doi: 10.1136/bmj.n71
- Putera, R. E., Mohamed, N. B., Wanaeloh, W., Fathani, A. T., Saputra, N., Valentina, T. R., et al. (2025). A hybrid model of disaster-resilient schools in Padang City, Indonesia. *Int. J. Safety Secur. Eng.* 15, 2541–2549. doi: 10.18280/ijse.151211
- Putri, T. E., Darmawan, P., and Heeks, R. (2023). What is fair? The experience of Indonesian gig workers. *Digit. Geograp. Soc.* 5:100072. doi: 10.1016/j.diggeo.2023.100072
- Rosenblat, A., and Stark, L. (2016). Algorithmic labor and information asymmetries: a case study of Uber's drivers. *Int. J. Commun.* 10, 3758–3784.
- Ruyter, A. D., Brown, M., and Burgees, K. (2019). Gig work and the fourth industrial revolution: conceptual and regulatory challenges. *J. Int. Aff.* 72, 37–50.
- Sarker, M. R., Taj, T. A., Sarkar, M. A. R., Hassan, M. F., McKenzie, A. M., Al Mamun, M. A., et al. (2024). Gender differences in job satisfaction among gig workers in Bangladesh. *Sci. Rep.* 14:17128. doi: 10.1038/s41598-024-68327-5
- Security (2020). Government of India (2020). The Code on Social Security, 2020 (Act No. 36 of 2020). Ministry of Law and Justice.
- Shade, L. R. (2018). Hop to it in the gig economy: the sharing economy and neo-liberal feminism. *Int. J. Media Cult. Polit.* 14, 35–54. doi: 10.1386/macp.14.1.35_1
- Shockley, K. M., Shen, W., DeNunzio, M. M., Arvan, M. L., and Knudsen, E. A. (2017). Disentangling the relationship between gender and work-family conflict: An integration of theoretical perspectives using meta-analytic methods. *J. Appl. Psychol.* 102, 1601–1635. doi: 10.1037/apl0000246
- Srnicek, N. (2017). The challenges of platform capitalism: understanding the logic of a new business model. *Juncture*. 23, 254–257. doi: 10.1111/newe.12023
- Stanford, J. (2017). The resurgence of gig work: historical and theoretical perspectives. *Econ. Labour Relat. Rev.* 28, 382–401. doi: 10.1177/1035304617724303
- Stringhi, E. (2022). Addressing gendered affordances of the platform economy: the case of UpWork. *Internet Policy Rev.* 11. doi: 10.14763/2022.1.1634
- Sun, T., and Li, Y. (2024). Effect of algorithmic transparency on gig workers' and proactive service performance: a moderated chain mediation model. *Am. J. Indust. Bus. Manag.* 14, 462–491. doi: 10.4236/ajbm.2024.144024
- Tandon, A., and Sekharan, A. (2022). Labouring (on) the app: agency and organisation of work in the platform economy. *Gend. Dev.* 30, 687–706. doi: 10.1080/13552074.2022.2130515
- Tang, Y., Guo, P., Tang, C. S., and Wang, Y. (2021). Gender-related operational issues arising from on-demand ride-hailing platforms: safety concerns and system configuration. *Prod. Oper. Manag.* 30, 3481–3496. doi: 10.1111/poms.13444
- Turner, M., Holdsworth, S., Scott-Young, C. M., and Sandri, K. (2021). Resilience in a hostile workplace: the experience of women onsite in construction. *Constr. Manag. Econ. Routledge*. 39, 839–852. doi: 10.1080/01446193.2021.1981958
- Valentina, T. R., Putera, R. E., and Purnama, T. H. (2025). Harmonizing Pancasila principles with collaborative governance politics for sustainable waste management in Padang. *Int. J. Environ. Impacts*. 8, 865–875. doi: 10.56578/ijeio.80503
- van Doorn, N., Ferrari, F., and Graham, M. (2023). Migration and migrant labour in the gig economy: an intervention. *Work Employ. Soc.* 37, 1099–1111. doi: 10.1177/09500170221096581
- Vyas, N. (2021). ‘Gender inequality- now available on digital platform’: an interplay between gender equality and the gig economy in the European Union. *Euro. Labour Law J.* 12, 37–51. doi: 10.1177/2031952520953856
- Walby, S. (1990). *Theorising Patriarchy*. Cambridge, MA: Blackwell.
- Wiesböck, L., Radlherr, J., and Angélique Vo, M. L. (2023). Domestic cleaners in the informal labour market: new working realities shaped by the gig economy? *Soc. Incl.* 11, 262–273. doi: 10.17645/si.v11i4.7119
- Wood, A. J. and, Lehdonvirta V. (2019). *Platform Precarity: Surviving Algorithmic Insecurity in the Gig Economy*. New York City, NY: Society for the Advancement of Socio-Economics.

- Work Directive (2023). European Parliament and Council of the European Union (2024). Directive (EU) 2024/2831 of 23 October 2024 on improving working conditions in platform work. Official Journal of the European Union, L 2024/2831, 11 November 2024.
- Workers Rights Act (2022). Government of Ontario (2022). Digital Platform Workers' Rights Act, 2022, S.O. 2022, c. 7, Sched. 1 (enacted as part of the Working for Workers Act, 2022).
- World Bank. (2023). *Working Without Borders: The Promise and Peril of Online Gig Work*. Available online at: <https://ilostat.ilo.org/data/> (Accessed January 18, 2026).
- Yoo, H., Yang, M., Song, J. H., Yoon, J. H., Lee, W., Jang, J., et al. (2024). Investigation of working conditions and health status in platform workers in the Republic of Korea. *Saf. Health Work.* 15, 17–23. doi: 10.1016/j.shaw.2024.01.002
- Zhuang, C., Gu, T., Chung, H., Zhu, M., and Yonto, D. (2025). Spatial-temporal insights into gender gaps in East Asian ride-hailing: workload, efficiency, nighttime safety, and operational patterns. *J. Transp. Geogr.* 125:104213. doi: 10.1016/j.jtrangeo.2025.104213